

- 7 Insulin is transported around the body in the blood. The cells of the liver, muscle tissue and adipose (fat) tissue have receptors for insulin.

(a) State:

- the precise cellular location of the insulin receptors
- the type of biological molecule that forms an insulin receptor.

location

type of biological molecule

[2]

(b) Insulin stimulates the activity of the enzyme glycogen synthetase in liver cells.

Fig. 7.1 shows the activity of glycogen synthetase in liver cells for 210 seconds after glucose has been injected into the bloodstream.

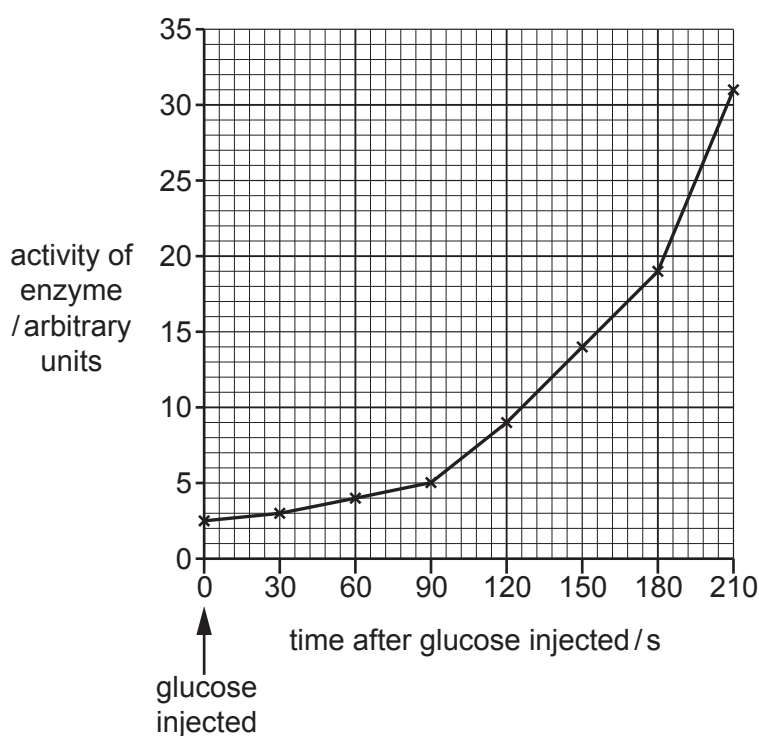


Fig. 7.1

- (i) Calculate the percentage increase in glycogen synthetase activity between 90s and 180s after glucose was injected into the blood.

Show your working and write your answer to the nearest whole number.

answer% [2]

- (ii) Suggest the role of glycogen synthetase in the regulation of blood glucose concentration.

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..... [1]

[Total: 5]